

Name Student: Tiffany Y. Chen

Internship or Thesis: Internship

Title project: A deeper understanding on human factors and ergonomics engineering in the Medical Device Design Process

(Abstract, 100-150 words)

ONWARD is designing products delivering targeted, programmed electrical stimulation of the spinal cord to restore movement and other functions in people with spinal cord injury, with the purpose of restoring life enjoyment in every way that matters to them. When designing medical devices, Human Factors engineering is essential in creating devices that are safe and effective to use. User Interface Interaction design shortcomings can lead to use errors, and eventually, lead to harm to the user in various degrees of severity (e.g., physical injuries). Besides, applying Human Factors from the beginning of the design and development process has a few benefits including shorter time to market, increased customer satisfaction, reduced demand for customer support, etc. In this presentation, I explain the different Human Factors Engineering activities in the design process with elaborations on my contributions, and how I applied what I learned from the master's program to this internship.

Name company or University: ONWARD Medical

(If applicable) External supervisor: Francesca Stradolini

UU Supervisor: Chris Paffen

Name Student: Hendrik Gunawan

Internship or Thesis: Internship

Title Project: The Art of Handling Careless Responders

The purpose of the reaction time check was to improve the attention of responders by using checks on the task at hand in online fear conditioning task. The fear conditioning task was a process of fear learning which presents responders with faces and auditory stimuli. Selective attention to the presented stimuli was essential as it influenced the fear conditioning experienced by the responders. In order to make sure participants had good attention on the fear conditioning task, check such as reaction time check was implemented. Other checks such as instructed questions & catch trials were considered for implementation in the experiment. However, it was feared an excessive use of checks would distract participants focus from the experiment.

Name company or University: Utrecht University

(If applicable) External Supervisor: none

UU Supervisor: Prof. dr. Joke Baas

Sasha Silberberg
Internship (12,5 EC)

*Improving the User Experience of the Teaching and Learning Collection: Usability Research via
User Interviews & Dashboard Design*

The aim of this internship was to evaluate user behavior and improve the user experience of the Teaching and Learning Collection ("the Collection"), an educational database created by Utrecht University's Centre for Academic Teaching & Learning ("CAT"). Five user interviews with Utrecht University educators were conducted in order to discover how users engage with the Collection in its current form and what kinds of changes can be implemented to make the Collection more valuable. A thematic analysis of interview transcripts unveiled both the strengths and weaknesses of the Collection's user experience. Suggestions for new features and improvements to the Collection's accessibility, messaging, and structure were made. In order to give the CAT increased and continuous oversight of the Collection's performance, a customized Google Analytics dashboard was also created. This dashboard highlights insights relevant to user engagement and website accessibility via metrics focused on user behavior and website traffic, respectively.

Organization: Utrecht University Centre for Academic Teaching & Learning
External Supervisor: Anne Geesink (a.m.geesink@uu.nl)
UU Supervisor: Jessica Heeman (j.heeman@uu.nl)

Name Student: Vesa Turjaka (2335174)

Internship or Thesis: Internship

Title project: Determinants of Engagement of Senior Residents with Dementia: An Explorative Study towards Unsupervised Purposeful Play

Name company or University: Tover B.V.

(If applicable) External supervisor: Claire Bernaards, PhD

UU Supervisor: Dr. Caroline Junge

Abstract

The purpose of the Unsupervised Purposeful Play (UPP) project is to explore the determinants of successful UPP in order to improve engagement of people living with dementia (PwD). UPP entails residents playing independently with the Tovertafel after it was switched on by healthcare personnel. The desirability (acceptance, willingness, etc.) and feasibility (from environmental factors towards game characteristics) of UPP were investigated. The results presented derived from literature studies, interviews and video-analyses. Unfortunately, testing was not feasible given the short time duration and COVID outbreaks in possible test locations. Recommendations and potential hypotheses are provided for the Researchers and Game Designers at Tover. Nevertheless, it was concluded that UPP might be desirable from an expert perspective, but not (yet) from a healthcare perspective. To increase the feasibility of UPP, advice is provided in regards to the environment, group compositions, game types and game characteristics. However, the aforementioned assumptions should be tested.

Keywords: Dementia, Seniors, Supervised/ Unsupervised Purposeful Play, Meaningful Engagement, Serious Games

Name Student: Alisso Andrikopoulou

Internship or Thesis: Internship

Title project: Avoiding fatigue: Healthy roster characteristics, countermeasures and the use of Biomathematical Models as evaluation tools.

Abstract: Fatigue is considered to be a major concern in the occupational field, due to its impact on the physical and mental health of employees and also on their working performance. Shift scheduling systems seem to play a vital role on the onset and progression of fatigue. Many costumers of INTERGO are companies where employees work at safety-critical positions and that mostly involve shift work (24/7). There is an urge for advice regarding effective rostering systems and that brings the company in need for up-to-date knowledge. The aim of this project is to provide INTERGO with an evidence-based advice regarding fundamental characteristics for healthy rosters. Bio-Mathematical models are used as evaluation tools, while at the same time a recommendation is to be given concerning the most effective BMMs based on certain criteria. Regarding its approach, the project is divided in two main sections, literature review and assessment of Bio-Mathematical Models using real rostering data provided by the company. The results will be presented at a report containing the final advice, but are yet to be defined, as well as the conclusions, as the project is still on progress.

Name company or University: Intergo

(If applicable) External supervisor: Richard van der Weide

UU Supervisor: David Terburg

More high schools than expected and relatively many have stopped using method H for the history classes in the lower grades in havo/vwo. To gain insights in the reasons for this occurrence, Mediatest has conducted six online interviews with history teachers of different high schools that have experienced using this method. The main finding is that the reasons for not using method X are based on the content and that competing methods are better aligned in this sense. This overall conclusion is based on three clusters of findings, namely 1) the content of method X is too complicated, 2) the structure of method X is too complex and 3) these difficulties with using method X are not improved in the latest edition.

Name Student: Olga Sagou

Internship or Thesis: Internship

Title project: Head Orientation and Gaze Direction of Experienced and Inexperienced Cyclists: A Pilot Study

Main purpose of this research was to find out whether experience in a task that is automated with practice, in this context bicycling, has any influence on visual behaviors. The field study consisted of 20 experienced and 20 inexperienced cyclists. They all followed the same route in the Utrecht University campus while wearing eye-tracking glasses by Pupil Labs. It was expected that experienced cyclists will rotate their heads and shift their gaze more than the inexperienced group, since they do not need to focus on the cycling task (which is automatized for them) and can freely observe their surroundings. Some results were in line with the research hypothesis, with the most notable difference being the number of head movement and average completion time variables. The gaze position of the inexperienced participants was slightly lower on the video images. This had as a result that the bike could be seen in many footages of the inexperienced group. This research is in its pilot phase, with a larger study to be conducted to resolve the manual coding of wearable eye-trackers.

Name company or University: Utrecht University

UU Supervisor: Dr. Ignace Hooge

Name Student: Jolijn Jansen

Internship or Thesis: Internship

Title project: Exploring nonverbal communication in dyadic puzzle solving

Social interactions and problem-solving strategies were explored in a dyadic mute turn-taking puzzle task. In the experiment dyads recreated figures by adding one tile per turn. The outcome had to be communicated without use of verbal communication or pointing. Non-verbal communication was recorded and transcribed, including gaze direction, body language and cues through tile placement. Most dyads successfully solved half of the trials and at least two of the last three trials. Gender, figure orientation, communication method, start of a strategy, spatial awareness and duration did not relate to dyad performance. However, dyads that solved the most trials in the least time mainly applied a similar strategy: using placement and orientation of tiles to communicate. Hence, dyads can solve a turn-taking puzzle task without verbal communication or pointing. Factors that influenced their performance could not be discerned, yet indirect communication through puzzle tiles was applied more successfully than other strategies.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Intuitively Improving Intuitive Neuromarketing

Student: Vivianne Streefkerk

Internship or Thesis: Internship

Title project: Intuitively Improving Intuitive Neuromarketing

Name company: Bloakes

External supervisor: Tycho Faber

UU Supervisor: Famke Asberg

Bloakes is an intuitive neuromarketing company that focuses on optimizing consumer decisions by measuring emotions with binary choice tasks and priming tasks. In September 2022, Bloakes has launched their new platform which they use to build and perform these tasks. One concern with the new platform is that during the data collection, many participants quit the test without finishing it. It was our hypothesis that the current practice block is too comprehensive and either confusing or boring participants. Therefore, a research has been performed to test whether the current practice block could be replaced with a shortened practice block or just verbal instructions. Results show that there were significantly less drop-outs with the new practice blocks compared to the current one. It has also shown that a shortened practice block is preferred to just verbal instructions, so people can get familiar with the platform and the test.

Name Student: Femke Verbeten (6434274)

Internship or Thesis: Internship

Title project: Establishing and Improving the User Experience of the Application

GebruikersInZicht

Since the target group of accessible UX-research are individuals with an impairment, it is of importance that the used methods are well tailored to the target group.

Therefore, the Accessibility Foundation designed a mobile application that is intended to provide questionnaires in an inclusive way. In the current study, the user experience of the application is investigated. This has been done by doing user experience research in which 8 participants with either a motor or visual impairments have tested the app on location and provided their experiences afterwards in an interview. A thematic analysis has shown that the participants experiences reveled 9 themes and 22 subthemes. A main positive point that participants have experienced was that they enjoyed the structure and simplicity of the app. Main negative points that participants have experienced were the unclarity of the buttons and not being able to answer questions when using a screen reader.

Name company or University: Accessibility Foundation

(If applicable) External supervisor: Aafke van Welbergen

UU Supervisor: Krista Overvliet

Name Student: Esmée Teinsma

Internship or Thesis: Internship

Title project: The effect of Inflation on Consumer Behavior towards Supermarket Purchases

Abstract

The current world situation has affected the increase in product prices in supermarkets. Since 2016, food insecurity has been increasing and, in addition, the war between Russia and Ukraine has caused food shortages and an increase in raw material and product prices. This has led to a high inflation rate in 2022. Due to inflation, food products in supermarkets in August 2022 were 18.5% more expensive than they were a year earlier. As prices have risen significantly, some consumers will have to adjust their spending behaviour to the new prices. The extent to which consumer purchasing behavior has changed depends on what drives consumers when making a purchase and the living situation in which the consumers are in. To investigate how inflation has affected consumer purchasing behavior in supermarkets in the Netherlands, a questionnaire was used to examine the changes in purchasing behavior of consumers since the increase in prices in supermarkets. The focus was on changes in product purchases, brand purchases, changes in purchased quantities and changes of supermarkets. Especially in households with children living at home, in less stable financial situations, and for consumers who pay attention to the price-quality ratio when buying a product, inflation has had a great effect on purchasing behavior in supermarkets. In these groups, less of certain products are bought, more of some products are bought, more private labels of products are bought, and some costumers have changed to different supermarket, since the prices have increased.

Keywords: consumers, inflation, product prices, purchasing behavior, supermarket

Name company or University: ESSENSOR

(If applicable) External supervisor: Sophia Eijnsman

UU Supervisor: Dr. Chris Klink

Workshop assignment (Arslaan Amjad, 6617905)

Abstract

Detecting visual field deficits at an early age is imperative as it can prevent any visual deterioration. However, assessing the visual field of children remains a challenging procedure as these require a certain degree of cognitive capacity and (non)verbal skills, which young children might not possess yet. This may pose a challenge when working with young children when trying to diagnose them with a visual deficit. It is therefore important to develop an objective tool that gives reliable results in the face of these challenges. Measuring the pupillary response may provide a solution for these challenges. Previous research has shown that the pupil constricts to various types of visual events, such as changes in color. (Strauch et al., 2022; Sperandio et al., 2018; Barbur, Harlow, & Sahraie, 1992). As the pupil size is affected by changes in color, it could be argued that this may be an objective indicator of awareness of these changes and can be used as a biomarker of color blindness. Therefore, the aim of this research will be to, first, explore whether pupil responses in the face of changes in color in the environment are indicative of color blindness, and second, whether the severity of color blindness can be determined when comparing these pupil size oscillations in a standardized environment. It is expected that people with color blindness will have weaker response to the changing colors than people who are not colorblind. To test this hypothesis, the pupil size is measured with an eyetracker while participants view dynamic stimuli that change in their color. This test is done with colorblind participants and control (non-colorblind) participants. Results from this test indicate that colorblind participants show a lower mean pupil amplitude response to colors their vision is weak to when compared to the control participants. In addition, the results show individual variance in the pupil amplitude response among colorblind participants. This could possibly indicate variance in sensitivity towards certain color spectrums, meaning there is a difference in the severity of their colorblindness. The results of this study provide more insight towards into the mechanisms of the eye and the pupil when watching (changing) colors. In addition, this study provides support in using the pupil response as an objective measure for detecting visual deficits, such as colorblindness. We hope that the presented results will stimulate further research in this topic and that it will lead to the development of an objective and reliable measurement method of visual field deficits in children.

Keywords: pupillometry, colorblindness, color

Name Student: Anneliek Heuvel

Internship or Thesis: Internship 12.5

Title project: Biodiversity, Behaviour and Barriers: Factors that Influence Citizen Engagement with Biodiversity

Abstract:

It has been shown that private gardens and urban green spaces play an important role in providing biodiversity within cities. As the world is experiencing biodiversity loss it is essential to gain insights into how urban biodiversity can be supported by citizens. Towards this goal this research investigates which factors influence behaviours contributing to biodiversity. This is done by a literature review and creation of a Societal Readiness survey. The survey data is collected in quantitative and qualitative form, some of which will be transformed to quantitative data by frequency analysis. The findings provide an overview of the factors that influence Dutch citizens' behaviour towards biodiversity conservation. Primary factors found include social norms, maintenance intensity, and nature connectedness. These results suggest that Dutch citizens may be supported to engage in behaviour that supports biodiversity through addressing social norms, removing maintenance barriers, and fostering nature connectedness. These insights can be used to advise governmental bodies and educational institutions.

Name company or University: Hogeschool van Amsterdam, Het lectoraat Psychology voor een Duurzame Stad

(If applicable) External supervisor: Dr. Loes Kreemers

UU Supervisor: Charlotte van de Wouw

Name Student: Pieter Slappendel

Internship or Thesis: Internship

Title project: Cognitive Biases applied in website design to increase conversion rates of ecommerce companies.

Abstract:

Over the past years, ecommerce has become increasingly more important in people's daily life. Ecommerce companies are continuously trying to grow their business by increasing conversion rate. Companies could make use of various cognitive biases in their website design to realize this. The current paper sought to answer the question: How can cognitive biases be used in website design to increase conversion rates in ecommerce companies? To answer this question, five cognitive biases are examined, and some are practically tested through A/B testing. Whereafter results are analyzed. This resulted in the formulation of ways in which cognitive biases can be practically applied to websites to increase conversion rate. An implication of the current research is that this is one of few scientific papers that links scientific literature about cognitive biases to practical applications for website design. Further research is thus recommended on this topic.

Key words: conversion rate | e-commerce | website design | cognitive biases

Name company or University: Brandfirm

(If applicable) External supervisor: Jorien van der Weerd

UU Supervisor: Isolde Maas

Name Student: Ivet Riba Ustrell

Internship or Thesis: Internship

Title of the project: Navigating the Changing Labour Market: Generational Trends at Havas Lemz

This report aims to help Havas Lemz understand the tension in the labour market due to changing perceptions of work. According to the UWV (Netherlands Employee Insurance Agency), 75% of professions are experiencing extreme scarcity of new employees, which is why it is crucial to understand the trends and opinions of the current workforce. Through an academic literature review and field research, including a survey of 68 employees and a workshop with employers, the report found gaps between generations and among employers and employees in how work is perceived in Havas Lemz. The report suggests its conclusions be implemented in the design of the company Employer Branding Proposition and actions to close generational gaps in order to attract new talent and retain current employees.

Name of the company or University: Havas Lemz

(If applicable) External supervisor: Michiel Cox

UU Supervisor: Suzanne Hut

Name Student: Janneke van de Graaf

Internship or Thesis: Internship

Title project: The use of illustrated associations in implicit association tasks

In most of their studies, neuromarketing bureau Bloakes uses implicit association tasks. This method asks participants to repeatedly choose between two words (associations) as quickly as possible. To keep up with the needs of their clients, Bloakes needs more associations. During my internship, I examined whether it is possible to couple illustrations with association words. This was done using a binary choice task of four blocks, in which an illustration was coupled with each association. When going through the blocks, the illustration was shown to the participants. Together with this illustration, two association words were shown, which were either the specific association word for that illustration or other association words from the same topic. Following this, the participants were asked to choose one of the two words that they felt that matched best with the illustration. The results showed that the coupled association words were chosen over the other association words of the same construct for all four illustrations in significantly the majority of time. This study demonstrates that it is possible to couple an illustration with a specific association word.

Name company or University: Bloakes

(If applicable) External supervisor: Marily Westendorp van Anken

UU Supervisor: Famke Asberg

Improving FMO's Evaluation Reports and Digital Library Performance

By Understanding the User Experience

Internship | Gregoria Frederika Mandias

Abstract

The Evaluation and Knowledge Management (EKM) team of FMO (Dutch Entrepreneurial Development Bank) has produced evaluation reports to evaluate impact investments made by the company. Currently, the evaluation reports are gathered in FMO's intranet digital library, named the Evaluation Library. The EKM team felt the urge to understand the user's experience better to improve the evaluation reports and library, therefore, we conducted mixed-method research to answer, "How to improve the user experience on FMO's Evaluation reports and library?". The research found that 69% respondents are not aware of the evaluation reports and library (non-users). Based on the insights, we recommend the EKM team to highlight the importance of the findings, not only to specific departments, but also to overall FMO team.

Keywords: User experience, evaluation reports, digital library, intranet.

Company name: FMO (Dutch Entrepreneurial Bank)

External supervisor: Violeta Bertolini

UU Supervisor: Daan van Rooij

Name Student: Nabila Talitha

Internship or Thesis: Internship

Title Project: Optimizing User Experience on Learning Video-Based Website Through User Satisfaction Questionnaire and Website Performance Score at Psyflix

The purpose of this study was to (1) gather feedback on user satisfaction with Psyflix and (2) analyze the performance of the website. According to a previous study conducted using Page Speed Insight, the site has an overall performance of 77. While the website received a good score, there is still room for improvement. Apart from that, user feedback is also a necessary factor in determining whether a website meets users' needs. This study is done quantitatively, participants (n=49) filled out an online survey about their satisfaction with using Psyflix. In the next phase, this study also used Page Speed Insight to gather data on website performance and analyze if there were any differences from the previous study. The research is done to inform the users' triggers, barriers, and satisfaction to other teams' and translates them into suggestions on how to enhance the website's user experience.

Keywords: user experience, user satisfaction, online learning website, healthcare professionals

Name of company or university: Psyflix

External supervisor: Hardy de Heer

UU supervisor: Charlotte L. van de Wouw

Utilizing Cognitive Load to Interfere with Memory Consolidation

We are constantly making memories, and the process of those memories being consolidated is automatic. However, research has shown that while it is constantly ongoing, it is not unstoppable. Interference with this process is possible through multiple means, but for this literature review the focus is on how cognitive load interferes with consolidation. Not all tasks that put a strain on our cognition are effective at causing interference, for example, n back tests and video games do not have a measurable interference effect on consolidation, triggering the recall of autobiographical memories, or spot the difference games do. A potential cause for this may be whether an interference task activates the same part of the brain that memory may be stored in, like if the memory is episodic, the interference task may be more effective if it also utilizes the medial temporal lobe. The hope is that the trends noticed in this literature review for episodic, procedural, and working memory, can be recreated for emotional memory, providing a new avenue of approach for cognitive behavioral therapy.

Is Genre still valuable to Game Producers: Notion, Strategies, and Pitfalls

Abstract

This report is an extended research based on the data of console game industry in the original internship project **Motivations for First-Party Complement Provision**. In this report we mainly interested in the notion of genre in video games, including literature research about what are genres of video games, why they are important, and exploring strategies that console game producers can take advantage of to decide in which genre they should produce their new game to maximize the profit. Using data from the US console-video game industry from 1995 to 2005, the size of the game development team has been shown to be positively correlated with the game's revenue, but distinguishing for different genres. Our findings also indicate that there are cyclical trending patterns between games' popularity and genre/gameplay which enables certain complements to have a higher probability of success at specific periods of time. In the last part, the report also demonstrates some pitfalls: boundary collapses, and underlying similarities, of defining games with existing categories of genres.

Internship Host: Tilburg University
External Supervisor: Xiaowei Zhang
UU Supervisor: Dr. Dennis Schutter

Name of Student: Farah Hamadeh

Internship or Thesis: Combined

Title: Mind the Gap: Investigating the Impact of Individual Differences in Working Memory Capacity on Multimedia Learning

Abstract:

The purpose of this literature review is to examine the impact of individual differences in working memory capacity (WMC) on multimedia learning. WMC is defined as the ability to control attention and retrieve relevant information from long term memory (LTM), however the precise nature of the components contributing to individual differences in WMC remain indeterminant. This review investigates the relationship between differences in WMC, attention control and learning strategies with the emphasis on cognitive theories of multimedia learning. The following electronic databases were accessed: ERIC, PubMed, Science Direct, JSTOR, PsychInfo and keywords such as “individual differences in WMC” , “multimedia learning” and “attentional control and WMC” were used to search for relevant articles. This study aims to highlight the significance of incorporating WMC into the design and implementation of multimedia instructional materials and to identify potential strategies for mitigating difficulties encountered by lower-WMC students. A number of gaps in individual differences in WMC in multimedia learning were identified after interpreting research results.

Name of University: Utrecht University

UU Supervisor: Dr. Margot van Wermeskerken

Enhancing Situation Awareness in Teleoperation Across Diverse Domains: A Literature Review

Shawn X. Veldhuis

Student number: 9648771

Supervisor: Chris P. Janssen

Master of Science program: Applied Cognitive Psychology

Component: Internship 7.5 ECT

In collaboration with: Deutsches Zentrum für Luft- und Raumfahrt, DLR Berlin

External Supervisor¹: Andreas Schrank, Andreas.Schrank@dlr.de

External Supervisor²: Fabian Hub, Fabian.Hub@dlr.de

Date: 21-11-2022

Utrecht University

The project that I will finish my internship with is a literature review to investigate and evaluate the current state of situation awareness in teleoperation across various domains. The review will focus on the different techniques and methods that have been implemented to enhance situation awareness in teleoperation, and how these techniques have been applied in different domains such as manufacturing, mining, agriculture, and transportation. The goal of this review is to identify gaps in the current literature and provide recommendations for future research in the field of teleoperation and situation awareness.

I will be writing mostly the introduction and methodology for this review.

Tasks I did so far during the internship:

- I created 3D vector icons for an augmented reality application using a 3D app. It was very difficult because I had no experience in this, but after a while I got the hang of it.
- I created questionnaires with partly HTML code. I had never done this before, so it was nice to learn more about HTML.
- I preselected papers for the literature review based on certain criteria. I had to read approx.. 1700 abstracts of papers.
- I translated a German guideline for eHMI to English and created an overview what components an eHMI guideline should have.
- And currently I am working on the methodology for the literature view.

Name Student: Ebelien Min – 6281109

Internship or Thesis: Internship

Title Project: Signalling Task Demand

The overall aim of this research was developing a model to predict the task demand for train signallers in England. A part of this project consists of validating said model. The validation will be performed with the use of trials in England, in which the predicted task demand will be compared with the task demand to be assessed with the use of eye tracking as an objective measure, and observations and questionnaires as a more subjective measure. A test measurement with eye tracking was performed in December and has been analysed. Analysis of the eye tracking data was conducted with the help of replaying the raw data, creating heatmaps, gaze patterns and looking at different metrics of defined Areas of Interest (AOI). The eye tracking was proven to be useful for the validation of the hypothesized timeline, and sequence in which the signaller tasks were carried out.

Word Count: 147

Name company or University: Intergo B.V.

(If applicable) External supervisor: Melcher Zeilstra

UU Supervisor: David Terburg

Name Student: Juul van den Boogaard

Internship or Thesis: Internship

Title project: The Determinants of Engagement of Seniors with Dementia – An Explorative Study towards Unsupervised Purposeful Play

ABSTRACT

The aim of the Unsupervised Purposeful Play (UPP) project is to explore the determinants of successful UPP in order to improve Engagement of people living with dementia (PLWD) in residential care. UPP entails that residents play independently with the Tovertafel after a residential worker has switched on the Tovertafel. **Desirability** (e.g., acceptance, willingness, etc.) and **feasibility** (from environmental factors towards game characteristics) of UPP were investigated. Presented results were derived from (a) literature studies, (b) interviews, and (c) video-analysis. Unfortunately, testing was not feasible given time and COVID-19 restrictions. Recommendations and potential hypothesis are provided for ResDes (Tover's Research and Design Department). It was concluded that UPP might be desirable from an expert perspective, but not (yet) from a healthcare perspective. To increase feasibility of UPP, advice is provided regarding preferable environments, group compositions, game types, and game characteristics. Nevertheless, aforementioned assumptions should be tested.

Keywords: Dementia, Seniors, Supervised/ Unsupervised Purposeful Play, Meaningful Engagement, Serious Games

Name company or University: Tover B.V.

(If applicable) External supervisor: Claire Bernaards (PhD)

UU Supervisor: dr. Caroline Junge

UU colleague: Vesa Turjaka (please note that we worked together during our internship)

Name student: Xi Zhang

Internship or thesis: internship

Title project: Datafication of human consumption behavior using immersive virtual reality

Virtual Reality (VR) offers unprecedented opportunities to understand human cognition and behaviour beyond any research environment. Immersing users in realistic yet controlled simulation scenarios creates an authentic testbed for proactive behaviour interventions and support.

This internship aims to gain experience in cutting-edge technologies related to behaviour research. As the intern research assistant, I was expected to learn knowledge of the Unity game engine and use it to help to create a digital canteen, where high-fidelity food models with different calorie densities will be placed for users to view and interact with. And I was expected to assist with human-subject data collection using the advanced VR headset -Varjo aero.

During this internship project, I have read through and written a literature review on 14 published articles from 2014 into 2022, which have given me the appropriate understanding and knowledge of the current developments and applications of VR, AR, and XR, especially in psychology experiments field; I finished the official tutorial courses of Unity game engine and also learned some operation skills from my supervisor and colleagues, which allowed me to import the raw models made from WANDERlab by Unreal Engine and Blender to Unity (converted textures and materials to formats that are accepted by Unity) and incorporate the Varjo eye tracking SDK, as well as I have independently built a panel to show the real-time eye-tracking data in the digital canteen interactive interface; I gained the data collection methods, such as collecting locomotions from the frequently refreshed (x,y,z) orientation coordinates of the headset, and collating eye movement data from changes in gaze position and pupil diameter.

This project experience has improved my research skills and provided me with practical technology skills for my subsequent thesis. And the results of this project can demonstrate the potential and pitfall of VR as a promising research tool for investing in human behaviour and facilitating the evidence-based design of eating behavior interventions.

Name company or university: Wageningen University&Research

External supervisor: Jiayan Zhao

UU supervisor: Leon Kenemans

Name Student: Yanna Litowsky



Internship or Thesis: Internship (12.5 ECTS)

Title project: Challenges of AI for wellbeing and mental health: Emotional expertise and granularity

Humans possess emotional expertise and granularity, in varying levels, to navigate life effectively. Research shows that individuals with high emotional expertise and granularity tend to have better wellbeing and health. As technology advances, there is increasing interest in developing artificial intelligence (AI) to support human wellbeing and mental health, such as therapy chatbots. However, the use of AI in this context raises concerns, such as the hypothesis that AI needs to possess emotional expertise and granularity to effectively align with and support humans. This research investigates the importance of emotional expertise and granularity of AI for supporting wellbeing and mental health. This project contributes to Dr. Lomas' study from TU Delft, through structured research and a reanalysis of its method for assessing the alignment of emotions expressed by generative AI. The research suggests that AI will likely need improved emotional expertise to adequately support mental health and wellbeing. The reanalysis presented improvement suggestions and implications for future AI systems designed for this purpose.

Keywords: emotional expertise, emotional granularity, artificial intelligence, generative models, AI alignment, emotions, wellbeing, mental health

Name of University: TU Delft | Faculty of Industrial Design Engineering

External supervisor: Willem van der Maden, Ph.D. candidate

UU supervisor: Dr. Robert Weijers MSc

Investigating the Roles of Intuition and Rationality behind the Creative Process with Generative Models. A Qualitative Study.

Xiyin Celine Chen, 2791307
Internship (27,5 EC)

Generative Models are algorithms that can evaluate and comprehend massive amounts of data to gain an understanding of the world. Their application has enormous potential and is becoming increasingly significant in fields such as medicine, education, and art. The use of Generative Models is a creative process that is divided into stages. They are dominated by either intuitive or rational thinking, often known as System 1 or 2 in the dual-process theory. The research investigates the role of intuition and rationality at various stages of creation with Generative Models. Expert users were interviewed to understand their creative process. Qualitative data analysis shows an interplay of intuition and rationality as well as the use of “rational expert intuition” and the relevance of this technology. We conclude that improving Generative Models for intuitive use would benefit both novice and expert users, as well as enhance the usability and efficiency of these models.

Faculty of Industrial Design Engineering, Technische Universiteit Delft
External Supervisor: Willem van der Maden, PhD Candidate
UU Supervisor: Robert Weijers, PhD